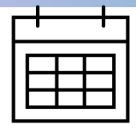
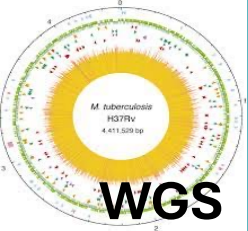


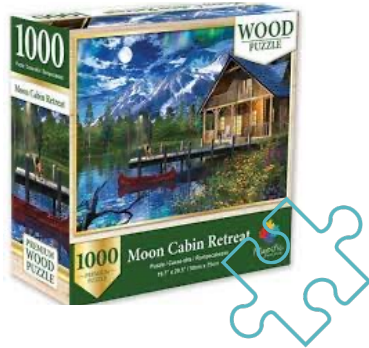
Decoding Drug Resistance in TB

From Cartridge to Genome

MOLECULAR ASSAYS FOR TUBERCULOSIS: increased levels of complexity

INFORMATION	<i>Mtb</i> /no <i>Mtb</i> RIF	<i>Mtb</i> /no <i>Mtb</i> RIF INH	<i>Mtb</i> /no <i>Mtb</i> 15 drugs Genotype	<i>Mtb</i> /no <i>Mtb</i> all drugs Genotype Transmission
TARGETS	MTB-specific 1 drug-target	MTB-specific 3 drug-targets	MTB-specific 18 drug-targets	4.4Mb All drug-targets
SAMPLE TYPE	Raw specimen	Treated specimen	Treated Specimen ***	Cultured isolate
MOLECULAR ASSAY	Simple NAATs  	Moderate complexity NAATs  	DEEPLEX tNGS  	WGS  

WGS DATA ANALYSIS APPROACHES



Reference mapping approach

- Known genome sequence of closely related organism
- Analysis is less resource-intensive
- Limited to known genomic regions



De novo assembly

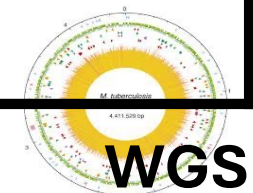
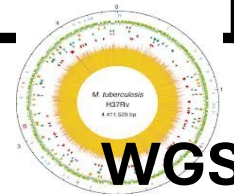
- Limited knowledge of genome sequence or closely related organism
- Analysis is resource-intensive
- Describe novel genomic structures
 - Species characterisation

ONT WGS ANALYSIS USING GALAXY

Mycobacterium tuberculosis clinical isolates

NGS DATA ANALYSIS TOOLS/PLATFORMS: increased levels of complexity

On/off line tools	Galaxy	Command line
• Automated • No installation • User friendly - GUI • Cloud-based • Internet connection needed* • Limited customizability • Runs CLI tools 'under the hood' • Potential data security issues	• Data analysis workflow platform • Hybrid interface • Ease of use of GUI • Flexibility of command line • Local or public server	• Highly customizable • Can be tedious • Needs computing recourses • Needs some training/familiarity with the software • Automation requires programming expertise



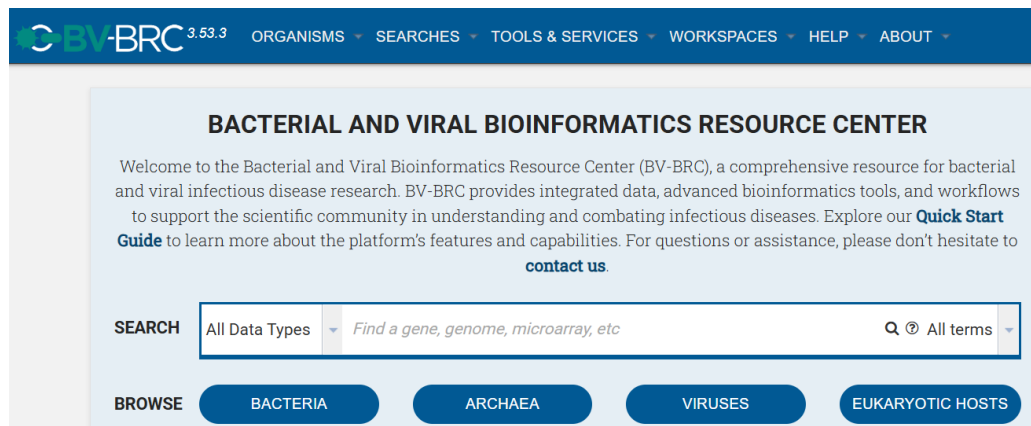
The logo icon consists of two horizontal dark grey bars of equal length, stacked vertically. Below the bottom bar is a yellow-to-white gradient bar that tapers to the right.

Galaxy

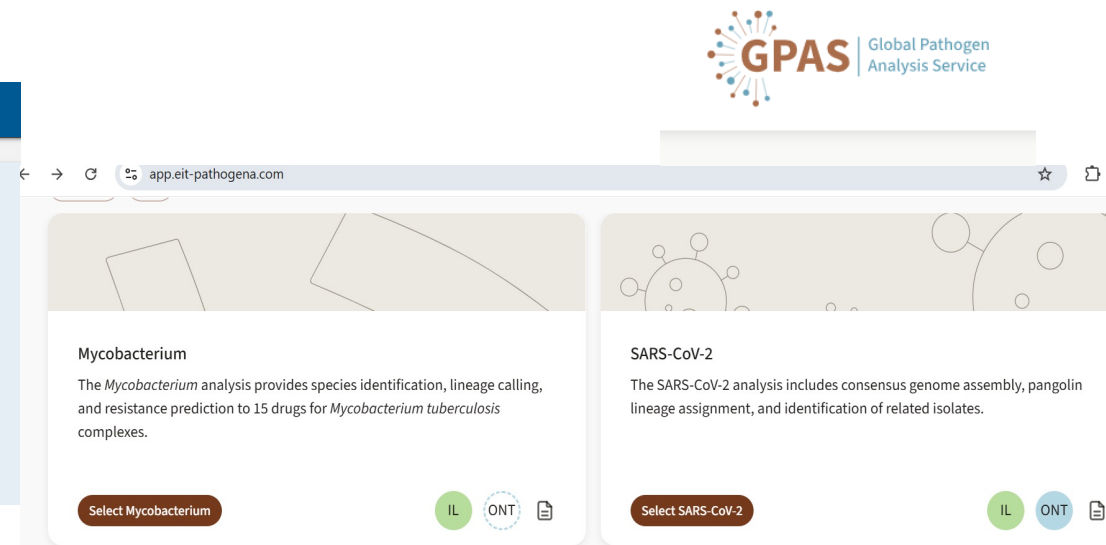
PROJECT

From programming languages to user-friendly genomics data analysis platform

<https://www.bv-brc.org/>



<https://app.eit-pathogena.com/>



From programming languages to user-friendly genomics data analysis platforms

The screenshot displays the Galaxy web interface. On the left, the 'Interactive Tools' sidebar is visible, with the 'Tools' icon circled in green. The 'RStudio' tool is highlighted with a green circle. The main panel shows the 'RStudio R 4.5.0 with Bioconductor 3.21' tool interface, including 'Tool Parameters' and 'Additional Options'. On the right, a workflow diagram titled 'Copy of TB Variant Analysis v1.1' is shown, illustrating a series of steps from '1: Reads' to '15: Filter failed datasets'.

Galaxy

Using 0% of 250.0 GB

Interactive Tools

Launch and manage interactive tools

R

interactive plotting tool for geo-referenced data

Qiskit Jupyter notebook interactive tool

Data Analysis Agent

Interactively chats using natural language to explore, visualize, and analyze your data

RStudio

R 4.5.0 with Bioconductor 3.21

PhysiCell Studio

Interactive tool to create, simulate, and visualize a multicellular model using PhysiCell

RStudio R 4.5.0 with Bioconductor 3.21

(Galaxy Version 4.5.0+3.21)

Run Tool

Tool Parameters

Include data into the environment - optional

No datasets available

switch to column select

Additional Options

Email notification

No

Send an email notification when the job completes.

Attempt to re-use jobs with identical parameters?

No

This may skip executing jobs that you have already run.

Run Tool

Help

The RStudio Interactive Tool in Galaxy provides a user-friendly interface for conducting statistical analysis, visualization, and scripting using the R programming language.

Copy of TB Variant Analysis v1.1 (imported from uploaded file) (imported from URL) shared t

1: Reads

2: Reference Genome

3: Minimum depth of coverage

4: Minimum variant allele frequency

5: Additional BWA-MEM options

6: fastp

7: samtools

8: Kraken2

9: snippy

10: mosdepth

11: QualiMap

12: Prepare list of SNPs for phylogenetics

13: VCF file to be filter

14: Coverage

15: Filter failed datasets

WGS ANALYSIS FOR *Mtb*

Comprehensive:

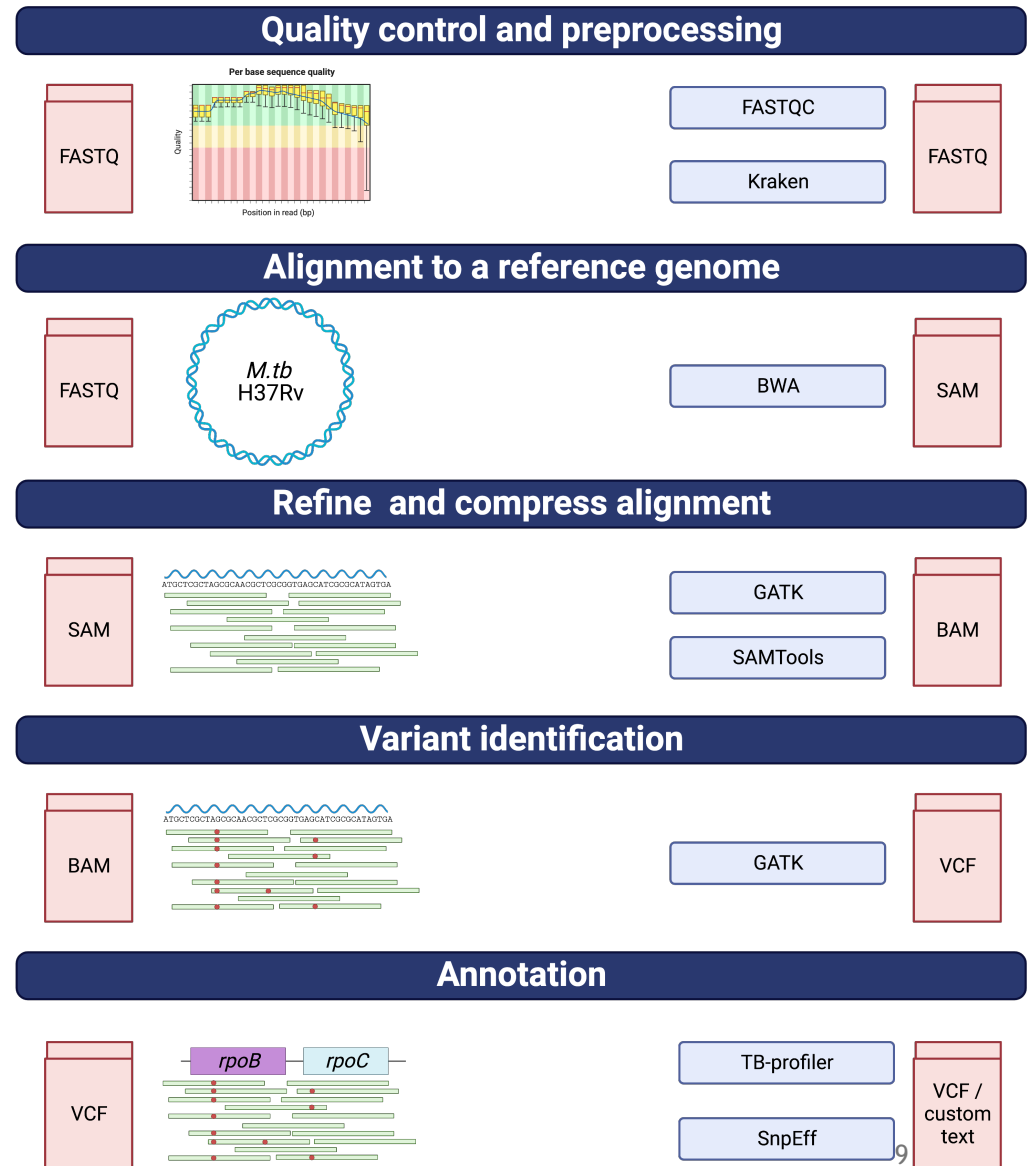
- QC
- Drug resistance
- Strain identification
- Genome-wide variants
- Transmission clusters
- Phylogeny

Focused:

TB Profiler

- Drug resistance
- Strain identification
- Basic stats for QC

• [10.1099/mgen.0.000689](https://doi.org/10.1099/mgen.0.000689)



TB-Profiler: web-based

The screenshot displays the TB-Profiler web interface. At the top, a dark navigation bar contains the links 'Home', 'Upload', and 'SRA data'. Below this, the 'TB Profiler' logo is prominently displayed, with the 'P' in 'Profiler' featuring a magnifying glass icon. A welcome message states: 'Welcome to the webserver of TB-Profiler - a pipeline which allows users to analyse *M. tuberculosis* whole genome sequencing data to predict lineage and drug resistance. Follow the instructions below to upload a new sample or view analysed runs.'

Under the heading 'How does it work?', a paragraph explains: 'The pipeline searches for small variants and big deletions associated with drug resistance. It will also report the lineage. By default it uses Trimmomatic to trim the reads, BWA (or minimap2 for nanopore) to align to the reference genome and GATK (open source v4) to call variants.'

The interface is divided into two main steps:

- Step 1: Profile your sample**
This section contains instructions: 'Upload your next generation sequencing data in **fastQ** format. You can upload one or two (forward and reverse) fastq files. When you upload your data, the run will be assigned a unique ID. Please take a note of this ID as you will need to find your results later. Batch upload of samples is also possible.' Below the text is a dark 'Upload' button.
- Step 2: View the results**
This section is currently empty, showing only the heading.

- Limited customisation
- One isolate at a time
- Access to 16,000 publicly available sequences
- User-friendly and no command line expertise needed
- Internet connection needed

<https://tbdr.lshtm.ac.uk/>

TB-Profiler: the command line

```
Usage: tb-profiler profile [-h] [--read1 READ1] [--read2 READ2] [--bam BAM] [--fasta FASTA] [--vcf VCF] [--platform {illumina,nanopore,pacbio}] [--db DB]
                           [--external_db EXTERNAL_DB] [--prefix PREFIX] [--csv] [--txt] [--text_template TEXT_TEMPLATE] [--docx] [--docx_template DOCX_TEMPLATE]
                           [--add_columns ADD_COLUMNS] [--dir DIR] [--depth DEPTH] [--af AF] [--strand STRAND] [--sv_depth SV_DEPTH] [--sv_af SV_AF] [--sv_len SV_LEN]
                           [--mapper {bwa,minimap2,bowtie2,bwa-mem2}] [--caller {bcftools,gatk,freebayes,pilon,lofreq}] [--calling_params CALLING_PARAMS]
                           [--kmer_counter {kmc,dsk}] [--suspect] [--spoligotype] [--update_phylo] [--call_whole_genome] [--snp_dist SNP_DIST] [--no_trim]
                           [--no_flagstat] [--no_clip] [--no_delly] [--threads THREADS] [--ram RAM] [--logging {DEBUG,INFO,WARNING,ERROR,CRITICAL}] [--temp TEMP]
                           [--version]
```

Input Options:

<code>--read1, -1 READ1</code>	First read file (default: None)
<code>--read2, -2 READ2</code>	Second read file (default: None)
<code>--bam, -a BAM</code>	BAM file (make sure it has been generated using the H37Rv genome (GCA_000195955.2)) (default: None)
<code>--fasta, -f FASTA</code>	Fasta file (default: None)
<code>--vcf, -v VCF</code>	VCF file (default: None)
<code>--platform, -m {illumina,nanopore,pacbio}</code>	NGS Platform used to generate data (default: illumina)
<code>--db DB</code>	Mutation panel name (default: tbdb)
<code>--external_db EXTERNAL_DB</code>	Path to db files prefix (overrides <code>--db</code> parameter) (default: None)

Output Options:

<code>--prefix, -p PREFIX</code>	Sample prefix for all results generated (default: tbprofiler)
<code>--csv</code>	Add CSV output (default: False)
<code>--txt</code>	Add text output (default: False)
<code>--text_template TEXT_TEMPLATE</code>	Jinja2 formatted template for output (default: None)
<code>--docx</code>	Add docx output (default: False)
<code>--docx_template DOCX_TEMPLATE</code>	Supply custom template for <code>--docx</code> output (default: None)
<code>--add_columns ADD_COLUMNS</code>	Add additional columns found in the mutation database to the text and csv results (default: None)
<code>--dir, -d DIR</code>	Storage directory (default: .)

- Flexible
- Fully customisable
- Parallel processing
- Internet only needed for initial installation and downloading publicly available genomes
- Updated version immediately available

MTB ONT WGS analysis on Galaxy with TBProfiler

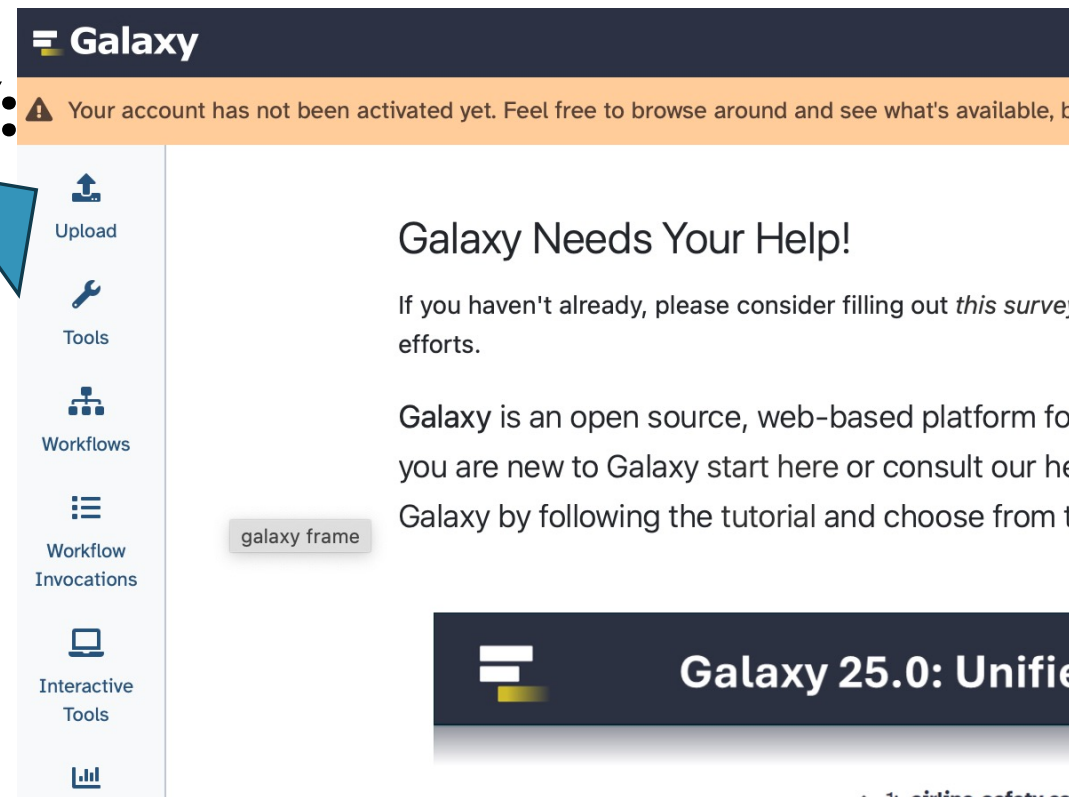
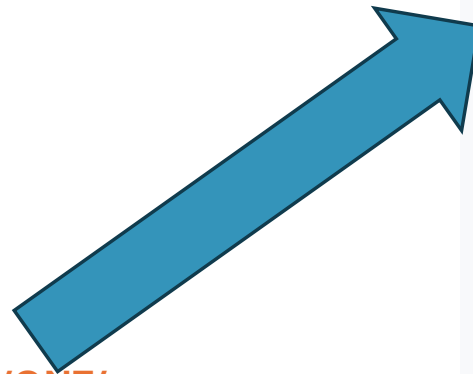
- Base calling has been done: FAST5 → FASTQ

- **UPLOAD DATA TO GALAXY:**

ONT FASTQ file

`/usr/local/share/MACMA/tbprofiler/ONT/`

ERR9030283.fastq.gz



UPLOADED DATA

The screenshot displays the Galaxy web interface. The top navigation bar includes the Galaxy logo, a grid icon, a graduation cap icon, a hexagon icon, a question mark icon, a storage usage indicator 'Using 0% of 250.0 GB', and a user profile 'anz.dippenaar'. The left sidebar contains icons for Upload, Tools, Workflows, Workflow Invocations, Interactive Tools, and a bar chart icon. The main content area is divided into three panels. The left panel, titled 'All Tools', lists categories like 'Get Data', 'Send Data', 'Collection Operations', 'GENERAL TEXT TOOLS' (with sub-items 'Text Manipulation', 'Filter and Sort', 'Join, Subtract and Group', 'Datamash'), and 'GENOMIC FILE MANIPULATION'. The middle panel, titled 'Search Datasets', has a search bar with 'search tools' and a table with columns 'Name', 'Tags', 'History', 'Extension', and 'Updated'. It displays two messages: 'No matching entries found for:' and 'No datasets found.'. The right panel, titled 'History', shows 'search datasets' and 'Unnamed history' with a total size of '274 MB'. It lists two datasets: '2: OS00120406_S3_L001_R2_001.fastq.gz' and '1: OS00120406_S3_L001_R1_001.fastq.gz'. A large blue arrow points from the 'No datasets found' message to the dataset list in the History panel.

Galaxy

Using 0% of 250.0 GB | anz.dippenaar

Upload

Tools

Workflows

Workflow Invocations

Interactive Tools

All Tools

search tools

Get Data

Send Data

Collection Operations

GENERAL TEXT TOOLS

Text Manipulation

Filter and Sort

Join, Subtract and Group

Datamash

GENOMIC FILE MANIPULATION

Search Datasets

Name	Tags	History	Extension	Updated
No matching entries found for:				
No datasets found.				

History

search datasets

Unnamed history

274 MB

2: OS00120406_S3_L001_R2_001.fastq.gz

1: OS00120406_S3_L001_R1_001.fastq.gz

PRACTICAL USING GALAXY

- STEPS FOR ANALYSIS:

- 
- Step 1: Search for TB-profiler in the Galaxy toolbox
 - Step 2: Customise the tool parameters
 - Step 3: Run TB-profiler on **one strain**
 - ...wait for the results
 - Step 4: Interpret the results

CUSTOMISING TB-Profiler PARAMETERS

- Choose the tool parameters appropriate to the analysis needed:
 - **Platform:** Nanopore
 - **Input File Type:** Single Fastq
 - **Output format:** text (the report in text format is more detailed compared to the slimmed-down, easy-to-interpret PDF report)
- The minimum depth and reporting allele frequencies can be customised under “Select advanced options” – ‘Yes’

TB-Profiler - GALAXY

- Set parameters
- Select dataset
- “Run Tool”
- Wait...

UNDERSTANDING TB-Profiler OUTPUTS

- Strain type
- Drug resistance profile
- Median depth
- Resistance to individual drugs

```
TBProfiler report
=====

The following report has been generated by TBProfiler.

Summary
-----
ID: strainA
Date: Thu Sep  7 06:31:41 2023
Strain: lineage4.3.2.1
Drug-resistance: MDR-TB
Median Depth: 71

Lineage report
-----
Lineage Estimated Fraction      Spoligotype      Rd
lineage4         1.000    LAM;T;S;X;H      None
lineage4.3       1.000    mainly-LAM        None
lineage4.3.2     1.000    LAM3              None
lineage4.3.2.1   1.000    LAM3              RD761

Resistance report
-----
Drug      Genotypic Resistance      Mutations
Rifampicin      R      rpoB p.Thr400Ala (0.94), rpoB p.Ser450Leu (0.96)
Isoniazid       R      katG p.Ser315Thr (0.93)
Ethambutol      R      embB p.Met306Ile (0.82)
Pyrazinamide    R      pncA p.Lys96Thr (0.91)
Streptomycin    R      rpsL p.Lys88Gln (0.90)
Fluoroquinolones
Moxifloxacin
Ofloxacin
Levofloxacin
Ciprofloxacin
Aminoglycosides
Amikacin
Capreomycin
Kanamycin
Cycloserine
Ethionamide
Clofazimine
Para-aminosalicylic_acid
Delamanid
Bedaquiline
Linezolid
```

TB Profiler for multiple samples (commandline)

- Look at Excel summary on VM:
/usr/local/share/MACMA/tbprofiler/

EXAMPLE_TB-profiler_summary_20230907

What kind of raw data do we get?

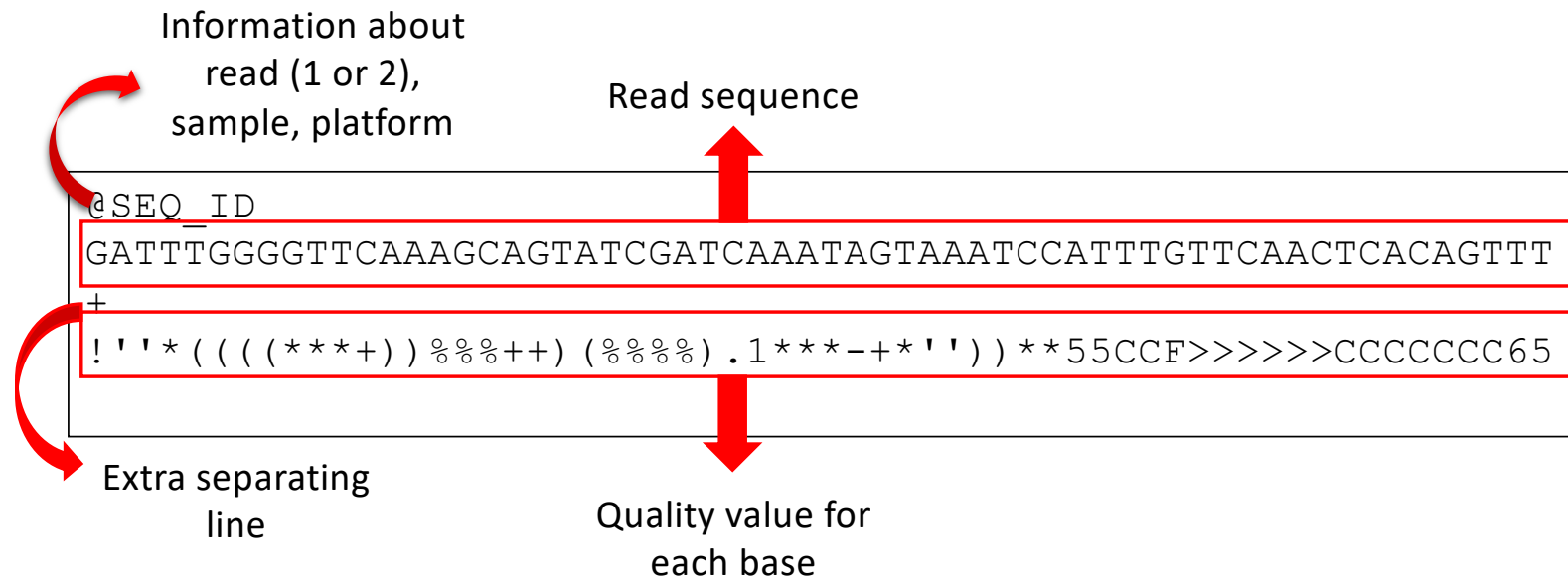


from Illumina sequencers:

- FastQ format: https://en.wikipedia.org/wiki/FASTQ_format

```
@SEQ_ID
GATTTGGGGTTCAAAGCAGTATCGATCAAATAGTAAATCCATTGTGTTCAACTCACAGTTT
+
!''*((( (**+)) %%%++) (%%%) .1***-+*'')) **55CCF>>>>>CCCCCCC65
```

FastQ format



Phred Quality values

- The quality score of a base, also known as a Phred or Q score, is an integer value representing the estimated probability of an error, i.e. that the base is incorrect. If P is the error probability, then:
 - $P = 10^{-Q/10}$
 - $Q = -10 \log_{10}(P)$
 - Translating to:
 - Q of 30 = error probability of 1 in 1000
 - Q of 20 = error probability of 1 in 100